# Week 5 Milestone 2: Benchmark

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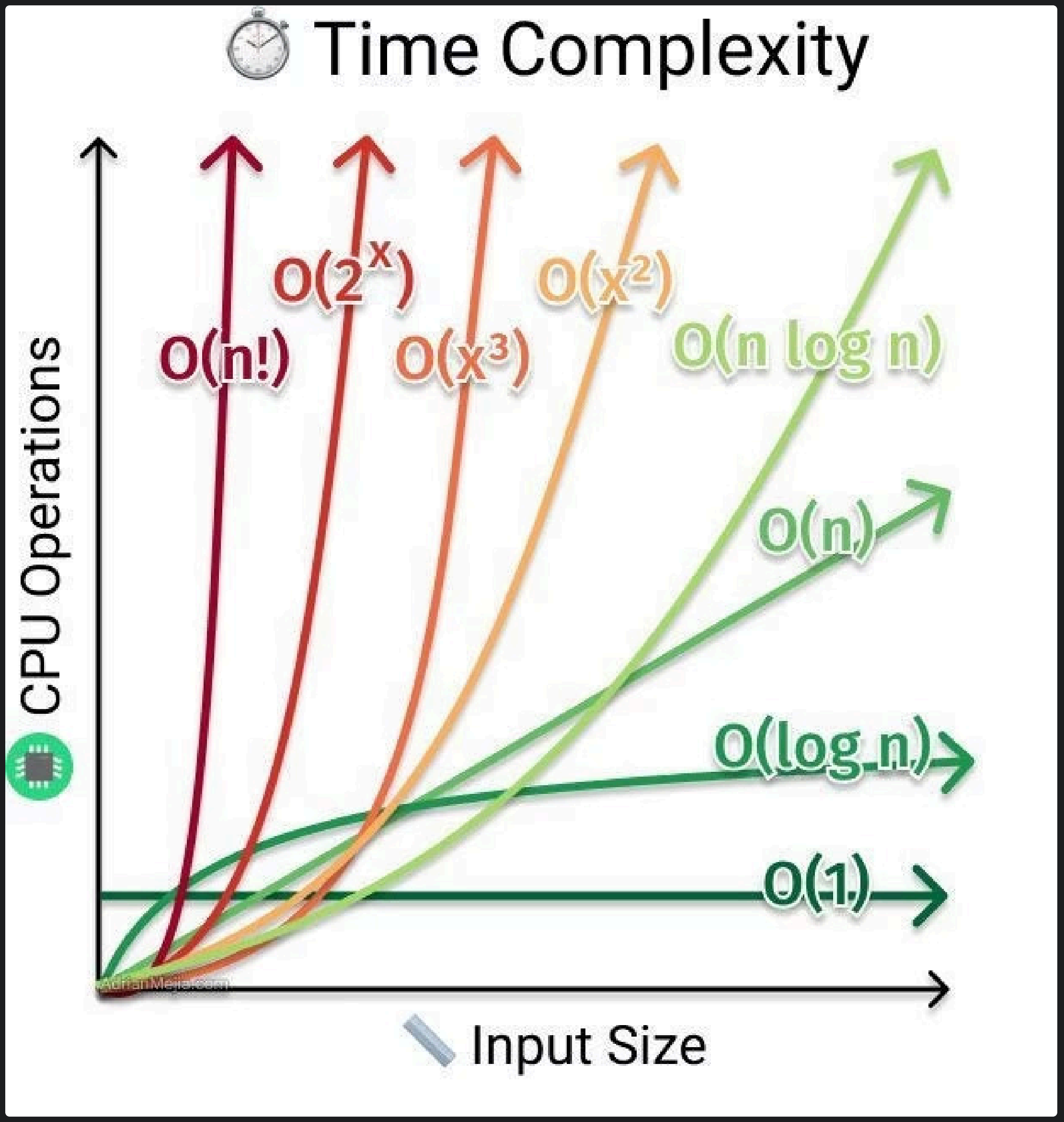
# Why these two new problems

| Problem | LLM's natural attempt | Why it fails | What the fix looks like |
| --- | --- | --- | --- |
| Two Sum Exists | $O(n^2)$ nested loops | TLEs at $n = 50\,000$ | $O(n)$ hash set / single pass |
| Fibonacci | $O(2^n)$ naive recursion | TLEs at $n > 35$ | $O(n)$ DP or memoization |

# Correct is not enough

| Type | Meaning | Explanation |
| --- | --- | --- |
| AC | Accepted | Correct AND fast AND small |
| TLE | Time Limit Exceeded | Correct but too slow |
| MLE | Memory Limit Exceeded | Correct but uses too much RAM |
| WA | Wrong Answer | Incorrect output |
| RE | Runtime Error | Crashed / Error |

# Quick Talk with O(n)



# TLE, MLE, and the judge

## [Naive Approach] Generating all Possible Pairs - $O(n^2)$ time and $O(1)$ space

```
for i in range(len(nums)):
    for j in range(i + 1, len(nums)):
        if nums[i] + nums[j] == target:
            return True
return False
```

# TLE, MLE, and the judge

## [Better Approach 1] Sorting and Binary Search - O(n × log(n)) time and O(1) space

```
seen = set()
for num in nums:
    if target - num in seen:
        return True
    seen.add(num)
return False
```

# Problem reveal + harness walkthrough